SECOND ANNUAL ECC
UNDERGRADUATE
MATHEMATICS COMPETITION

April 10, 1999

Time control: Three hours

To the team members: With any luck you will find these problems fun as well as challenging. You should probably not expect to work all ten of them in the allotted time. Each problem counts 10 points. While partial credit will be given for significant progress or for significant partial solutions, a thorough job on a few of them will be better than some exploratory work on all, so try to pick the ones which are most appealing to you and on which you think you can make some significant progress.

NO BOOKS, NOTES, CALCULATORS, COMPUTERS OR NON-TEAM-MEMBERS may be consulted.

Each team may submit one solution to each problem. Think of your solution as an essay; a logical argument which makes clear why your answer to the question is correct, or why the assertion whose proof is called for in the problem is true.

PLEASE BEGIN EACH PROBLEM ON A NEW SHEET OF PAPER. Team identification and problem number should be clearly given at the top of each sheet of paper.

1. Multiples of 11.

Prove that $2^{1999} + 5$ is a multiple of 11.

2. Pattern in squares.

Note that $2^2 + 3^2 + 2^2 3^2 = 7^2$ and $5^2 + 6^2 + 5^2 6^2 = 31^2$. Find a general pattern of which these are special cases, and prove it.

3. Any rationals here?

Given a positive integer n, let y_n be the decimal formed by concatenating the digits of 1^n, 2^n, 3^n ... and placing the decimal point at the left end. Thus, for example,

$$y_1 = 0.1234567891011\ldots,$$

$$y_2 = 0.1491625364964\ldots,$$

$$y_3 = 0.182764125216\ldots,$$

etc. Are there any integers n for which y_n is a rational number?

4. Which is greater?

Determine, with proof, which number is greater:

$$\left(\frac{1}{3}\right)^{1999} + \left(\frac{2}{3}\right)^{1999} \quad \text{or} \quad \left(\frac{1}{\sqrt{2}}\right)^{1999} + \left(1 - \frac{1}{\sqrt{2}}\right)^{1999}?$$

5. Approximating an integral.

(a) Find the points x_1 and x_2 so that the formula

$$\int_0^1 f(x)dx = f(x_1) + f'(x_2)$$

yields exact results for polynomials of degree ≤ 2. (As usual, f' denotes the derivative of f.)

(b) Determine the error (in absolute value) in using the resulting formula for a third degree polynomial $f(x)$ with leading coefficient 1. (The leading coefficient is the coefficient of x^3.)

6. Estimating a sum.

Let

$$S = \sum_{k=2}^{100} \frac{k^2-2}{k!} = \frac{2}{2!} + \frac{7}{3!} + \frac{14}{4!} + \cdots + \frac{9998}{100!}.$$

Prove that $S < 3$.

7. Close to an integer.

Let $a_1, a_2, \ldots, a_{1999}$ be a sequence of 1999 real numbers. Prove that there exists a sum of consecutive terms in the sequence which differs from an integer by less than .001.

8. Lots of nines.

A sequence (a_n) is defined recursively by $a_0 = 9$ and for integers $k \geq 1$,

$$a_{k+1} = 3a_k^4 + 4a_k^3.$$

Prove that in the decimal expression for the term a_{10}, more than 1000 of the digits are nines.

9. Roots of a cubic.

(a) Given that the equation $ax^3 + bx + c = 0$ with $a \neq 0$ and $b \neq 0$ has a double root r, show that $r = -3c/2b$.

(b) Show further that if the coefficients a, b and c are rational (and the equation has a double root), then

$$\sqrt{\frac{-b}{3a}}$$

is rational.

10. Minimization.

Let a, b, c, d, r and s be non-negative real numbers such that $a^2 + b^2 = r^2$ and $c + d = s$. Find the minimum possible value of $(a + d)^2 + (b - c)^2$.